



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE & TECHNOLOGY**

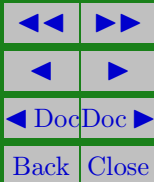
SCHOOL OF OPEN, DISTANCE AND eLEARNING

P.O. Box 62000, 00200

Nairobi, Kenya

E-mail: elarning@jkuat.ac.ke

ICS 2302 SOFTWARE ENGINEERING 1





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 10

Computer Security

Learning Outcomes

A study of this lesson should enable students to:

- Describe computer security



ICS 2302 Software Engineering 1

Data security is the means of ensuring that data is kept safe from corruption and that access to it is suitably controlled. Thus data security helps to ensure privacy. It also helps in protecting personal data.

10.1. Data Security Technologies

Disk Encryption

Disk encryption refers to encryption technology that encrypts data on a hard disk drive. Encryption is used to protect the message from the eyes of others. Disk encryption typically takes form in either software or hardware . It can be done in several ways by switching the characters around, replacing characters with others, and even removing characters from the message. These have to be used in combination to make the encryption

JKUAT: Setting trends in higher Education, Research and Innovation



ICS 2302 Software Engineering 1

secure enough, that is to say, sufficiently difficult to crack. Public key encryption is a refined and practical way of doing encryption. It allows for example anyone to write a message for a list of recipients, and only those recipients will be able to read that message.

Backups

Backups are used to ensure data which is lost can be recovered

Data Masking

Data Masking of structured data is the process of obscuring (masking) specific data within a database table or cell to ensure that data security is maintained and sensitive customer information is not leaked outside of the authorized environment.



Data Erasure

Data erasure is a method of software-based overwriting that completely destroys all electronic data residing on a hard drive or other digital media to ensure that no sensitive data is leaked when an asset is retired or reused.

Firewalls

Firewalls are systems which help protect computers and computer networks from attack and subsequent intrusion by restricting the network traffic which can pass through them, based on a set of system administrator defined rules.



Honey pots

Honey pots are computers that are either intentionally or unintentionally left vulnerable to attack by crackers. They can be used to catch crackers or fix vulnerabilities.

Intrusion Detection

Systems Intrusion detection systems can scan a network for people that are on the network but who should not be there or are doing things that they should not be doing, for example trying a lot of passwords to gain access to the network. viii.

File Integrity Monitors

File Integrity Monitors are tools used to detect changes in the integrity of systems and files.



Biometrics

Definition Here's a definition of biometrics: As it pertains to security, biometrics is an emerging field of technology devoted to identification of individuals by employing biological traits, such as those used for iris scanning, fingerprinting, face recognition, and others.

Biometric characteristics can be divided in two main classes:

- Physiological are related to the shape of the body. Examples include, but are not limited to fingerprint, face recognition, DNA, hand and palm geometry, iris recognition, which has largely replaced retina, and odor/scent.
- Behavioral are related to the behavior of a person. Examples include, but are not limited to typing rhythm, gait, and voice.



Brief description of the various types of Biometrics

- Fingerprints use patterns found on the fingertip some use optical images, while others use electronic field imaging. Ways to identify the fingerprint image include pattern matching, fringe patterns, ultrasonic and others.
- Face Recognition analyzes the characteristics of an individual's face. Facial recognition uses features of the face such as the location and position of the nose, outlines of the eyes, areas of the cheekbones, and the mouth.
- Hand Geometry measures the shape of the hand. Some hand scanners measure only two fingers, others use the entire hand for verification. These scanners look at the length, thickness, bone structure, curves and distance between the joints. This is accomplished by the reader cap-



turing high-resolution three dimensional images of the hand and comparing and verifying it to a database.

- Iris Scanning uses the unique characteristics of the human iris to identify an individual. Iris scanning begins with a picture. A camera with an infrared imager is used to illuminate the eye and capture a high resolution picture. The information about the iris is converted to an algorithm, which maps the iris's distinct patterns and characteristics.
- Retina Scanning involves the layer of blood vessels at the back of the eye. This technique involves a low level light reflected through an optical coupler to scan the unique identifiers of the retina. Retinal scanning requires the user to look into a scanner and focus on a specific point.
- Voice Authentication, not to be confused with voice recog-

JKUAT: Setting trends in higher Education, Research and Innovation



nition, is where software and hardware technology convert voice into text data that can be analyzed by voice response systems and other computers. Voice authentication uses the unique characteristics of an individual's voice for positive identification.

- Dynamic Signature Verification is the process used to identify an individual's hand-written signature by analyzing the shape, speed, velocity and pressure of the act of signing. The main difference between plain signature verification and dynamic signature verification is that the former is only the comparison of how the actual signature looks, while dynamic signature verification analyzes how the signature was made.
- Multi-modal biometrics combine different biometric appli-



cations. By integrating two or more biometric methodologies, these types of solutions meet the highest and most stringent security requirements.

10.2. Security Management

Security Management for networks is different for all kinds of situations. A small home or an office would only require basic security while large businesses will require high maintenance and advanced software and hardware to prevent malicious attacks from hacking and spamming.

Small homes

- A basic firewall like COMODO Internet Security or a unified threat management system.



ICS 2302 Software Engineering 1

- For Windows users, basic Antivirus software like AVG Antivirus, ESET NOD32 Antivirus, Kaspersky, McAfee, or Norton AntiVirus. An anti-spyware program such as Windows Defender or Spybot would also be a good idea. There are many other types of antivirus or antispyware programs out there to be considered.
- When using a wireless connection, use a robust password. Also try and use the strongest security supported by your wireless devices, such as WPA or WPA2.
- Use passwords for all accounts.
- Have multiple accounts per family member, using non-administrative accounts for day to day activities. Disable the guest account (Control Panel > Administrative Tools > Computer Management > Users).

JKUAT: Setting trends in higher Education, Research and Innovation



- Raise awareness about information security to children.

Medium businesses

- A fairly strong firewall or Unified Threat Management System
- Strong Antivirus software and Internet Security Software.
- For authentication, use strong passwords and change it on a bi-weekly/monthly basis.
- When using a wireless connection, use a robust password.
- Raise awareness about physical security to employees
- Use an optional network analyzer or network monitor.
- An enlightened administrator or manager.



Large businesses

- A strong firewall and proxy to keep unwanted people out.
- A strong Antivirus software package and Internet Security Software package.
- For authentication, use strong passwords and change it on a weekly/bi-weekly basis.
- When using a wireless connection, use a robust password.
- Exercise physical security precautions to employees.
- Prepare a network analyzer or network monitor and use it when needed.
- Implement physical security management like closed circuit television for entry areas and restricted zones.
- Security fencing to mark the company's perimeter.



ICS 2302 Software Engineering 1

- Fire extinguishers for fire-sensitive areas like server rooms and security rooms.
- Security guards can help to maximize security.

School

- An adjustable firewall and proxy to allow authorized users access from the outside and inside.
- Strong Antivirus software and Internet Security Software packages.
- Wireless connections that lead to firewalls.
- Children's Internet Protection Act compliance.
- Supervision of network to guarantee updates and changes based on popular site usage.



ICS 2302 Software Engineering 1

- Constant supervision by teachers, librarians, and administrators to guarantee protection against attacks by both internet and sneaker net sources.

Large Government

- A strong firewall and proxy to keep unwanted people out.
- Strong Antivirus software and Internet Security Software suites.
- Strong encryption.
- White list authorized wireless connection, block all else.
- All network hardware is in secure zones.
- All host should be on a private network that is invisible from the outside.



- Security fencing to mark perimeter and set wireless range to this.

10.3. Software Copyright

Software copyright, the relatively recent extension of copyright law to machine readable software. It is used by proprietary software companies to prevent the unauthorized copying of their software. It is also used by proponents of open source software to encourage the disclosure of improvements to source code.

Software Licensing Considerations

Software copyright is the legal protection that publishers and authors have regarding the use of their software. Software is covered by similar copyrights as those that apply to books etc.



ICS 2302 Software Engineering 1

When software is purchased, there is a software license that states the conditions for the use of the software. Use of the software in a way that is not allowed in its software license is illegal. In practice, copying software bought by someone else is likely to be an offence.

Different user licensing:

1. Single User License the software can be used on one computer only. This is the usual type of license.
2. Multi-user license an organisation may install the software on an agreed number of computers. A reduced fee is paid for each computer.



Freeware

Freeware refers to computer programs (software) that are given away free of charge. Freeware developers often still own the software copyright and can establish the terms under which it can be used and distributed. Users are not necessarily free to copy or distribute it further. Many freeware programs will include some information from the owner about how it can be used and any restrictions that may apply. You are legally bound to abide by any conditions the owner requires.

Shareware

Shareware distribution gives users a chance to try the software (program) for a specific period of time before deciding if they want to buy it or not. Shareware is basically “try before you



ICS 2302 Software Engineering 1

buy” software. After that, your trial period you have to either purchase the software and pay the owner, or uninstall the software from your computer system. Some shareware programs will continue to work after the trial period but will not be the last version of the software. Also, most shareware programs are limited in what they can do. E.g.: If it’s a game, it may only include the first level of a multi-level game. If it’s a business program, it may not allow you to save the files you are working with. Most shareware programs have a registration form that can be filled out and sent via Internet or printed out and mailed. Once you have sent the order form in, they send you the full version of the software, and register you at the same time, so that you will continue to get updated information on the software. If you register (purchase) the program, you will often receive a



ICS 2302 Software Engineering 1

password key to remove shareware 'nag' screens or you will be mailed a diskette/CD.

Site Licensing

Site Licensing enables you to buy additional copies of software like Microsoft Office for the office or lab at substantial discounts. You must own a full version of program and have the CD-ROM for the program you would like to get a site license for. Additional copies are then licensed at reduced rates. Site licensing gives you another Key Code or PIN Number for the software. You will use your original CD-ROM to install the software and enter the Key Code when prompted. You save money by not having to buy the complete package again and still everything will be legal.



ICS 2302 Software Engineering 1

Customer Eligibility: You can purchase this program if you are an Educational establishment (for teaching or educational research purpose only). Site licensing is used for schools that need multiple copies of the same software to outfit campus networks, computer labs, or classrooms. Site licensing offers 30-70% discounts off academic prices on single copies. Advantages:

- Significant cost savings.
- Standardization of software in the office and lab and therefore lowest cost of ownership.
- Peace of mind knowing that all the software is legal and licensed.



What are Patches?

If there's a problem with your software and you have an Internet connection, you can check out the software company's web site and see if they have a patch. A patch, generally speaking, is a way to update an existing piece of software on your computer to a new release.

What are Upgrades?

Software upgrade is a new version of a program or computer application, which is based on an older system but has been improved to fix problems or to add new features. Usually software upgrade consists of the files required to convert the older version to the newer version. It is usually meant to give added abilities to the program. No good program ever remains unchanged. Once



ICS 2302 Software Engineering 1

it starts to be used, there will be new ideas and suggestions as to how it can be improved. And, although it is tested thoroughly, sometimes a few small problems might be found, usually in unusual circumstances. Because of this, newer versions are usually produced, which eliminate the problems, and makes sure that the program continues to improve. Major new developments are sometimes referred to as 'Releases' (or 'Levels'). Versions that merely fine-tune the program are normally referred to as 'Upgrades'. Usually upgrades are given a number. A software program may go from 1.0 to 1.2 to 2.0 to 2.5 etc.

10.4. Software Installation and Configuration

Before using software you have to install it on your computer and choose particular settings so that it can be used on a given



ICS 2302 Software Engineering 1

hardware configuration. Installing a program means that you are adding a software program into the hard drive of the system. Some programs, usually windows, use the term setup when you actually go to add the software in your computer. This means you are asking the computer to setup the software so that you can use it.

How to install a program: Usually you just place the CD in the CD-ROM drive and it will ask if you want to install the software. Not all software is that easy. If the software does not have an auto install feature, read the documentation that came with the software for more specific instructions.



What is software registration?

Many times software registration is needed when software is installed. When new software is bought the user is often asked to register his copy with the company producing it. Registration is a technique used to prevent unauthorized use of software. The idea of copy protected software was created by software manufacturers who wanted to prevent people from making copies of software and giving them to friends and colleagues free of charge. Software registration is also when you order the software and downloading the full version.

Compatibility Considerations have to be made before installing the software on a computer. Some programs do not run on old computers due to hardware and software incompatibility.



Technical Support

Users may face problems due to program errors or lack of training so they might need help.

10.5. Software Piracy

Piracy is the illegal copying of software whether it's for personal use or for re sale. Software publishers lose possible income due to illegal copying of their products so they take measures to prevent it. Protected Software is supplied in a form design to be difficult to copy. It may have copy protection or use a key disk (see below). Unprotected Software has no special precautions against copying and depends on the legal authority of the software license. Copy Protection uses some physical changes to the data on the disk to prevent copy of the disk being made.



Installation disks

contain the software in encrypted form. The software is copied on the computer's hard disk before it can be used. The installation program may only allow the software to be installed a limited number of times, it may also make further copying difficult by embedding the owners name or computer serial number within the installation disks.



Revision Questions

Example . What is Confidential Data Storage

Solution:

Confidential data is any information you don't want others to obtain without your permission, including (but not limited to) your Social Security number, home address, phone numbers of friends/family/colleagues/students, your driver's license or bank account numbers, a list of your passphrases, your home address or phone numbers, your employee ID number, digital images, word documents containing personal text, etc. Most people store confidential data of some kind on their computers within Word files, address books, or application settings. ☐

EXERCISE 1.  What is copyright?



References and Additional Reading Materials

1. Hans van Vliet (2005). Software Engineering: Principles and Practice (2nd ed.). John Wiley & Sons, Inc. ISBN: 0-471-97508-7
2. Ian Sommerville, Pete Sawyer(2005). Requirements Engineering: A Good Practice Guide. ISBN: 0-471-97444-7. John Wiley & Sons, Inc
3. Gerald Kotonya, Ian Sommerville (1998). Requirements Engineering: Processes and Techniques. John Wiley & Sons, Inc. ISBN: 0-471-97208-8

Key disk is a floppy disk or CD-ROM disk used to install software which contains credits that are reduced by one software on each installation. It may also be a disk that is required every

ICS 2302 Software Engineering 1

time a program is run to ensure that is can only be used by one computer at a time



JKUAT SODeL
©2017

